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Part 1: Project Overview

Title: Graph-Based Text Mining for Research Paper Classification.

Dataset: The Cora Citation Network.

Objective: To develop a deep learning model capable of classifying research papers into one of seven categories by integrating both text content and citation relationships.

Part 2: Methodology

- **Intelligent Text Mining:** We used the bag-of-words features (1433 features per paper) as the primary input for our nodes.
- **Link Analysis & Graph Learning:** Instead of treating papers as independent data points, we represented them as a **Graph**. Citation links were treated as **Edges**, allowing the model to learn from the "network structure."
- **Graph Neural Networks (GNN):** We implemented a **Graph Convolutional Network (GCN)**. This architecture uses "Message Passing" to aggregate features from neighboring nodes, effectively capturing the context of each paper.

Part 3: System Architecture

1. **Input Layer:** Receives a feature matrix of size (2708 $\times$ 1433).
2. **GCN Layer 1:** Consists of 16 hidden units. It transforms the raw text features into a higher-level representation by neighborhood aggregation. We used **ReLU** activation and **Dropout** (rate=0.5) to prevent overfitting.
3. **GCN Layer 2:** Maps the hidden representation to the 7 output classes.
4. **Output Layer:** Applies a **Log-Softmax** function to determine the probability of each class.

Part 4: Discussion of Results

- **Final Accuracy:** The model achieved an impressive accuracy of **80.50%** on the test set.
- **Convergence:** The loss decreased from **0.2652** to **0.0362**, showing that the **Adam Optimizer** and a learning rate of **0.01** were highly effective for this dataset.
- **Scientific Conclusion:** The high accuracy confirms that citation links provide crucial information that text alone might miss. This proves the effectiveness of **Relational Deep Learning** in structured text mining.

Part 5: Execution Steps

1. **Environment Setup:** Ensure Python 3.10 is installed.
 2. **Install Dependencies:** Run `pip install -r requirements.txt` to install PyTorch, Torch-Geometric, and Pandas.
 3. **Data Placement:** Ensure the CSV files are located in the `/data` directory.
 4. **Run the Model:** Execute `python main.py` to start the training and evaluation process.
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